

TAMILSELVAN G

Full Stack Software Engineer | Angular | .NET Core | AI / RAG | SQL Server | Redis | AWS

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PROFESSIONAL SUMMARY

Full Stack Software Engineer with 4+ years of experience building and maintaining enterprise web applications. Skilled in Angular (14-20), ASP.NET Core (C#), SQL Server, Docker, and Redis, with experience improving application performance, backend efficiency, and production stability. Hands-on experience in REST API development, JWT authentication, SignalR, AWS S3, CI/CD, Clean Architecture, and Agile/Scrum development. Also experienced in RAG and LLM integrations using API-based services and Ollama.

TECHNICAL SKILLS

Frontend: Angular 14-20, TypeScript, JavaScript (ES6+), RxJS, HTML5, CSS3, Tailwind CSS, Bootstrap, Lazy Loading, ApexCharts, TinyMCE, Angular Material, Responsive UI Development, State Management, Reactive Forms.

Backend: ASP.NET Core, C#, Web API, RESTful Services, SignalR, JWT Authentication, Entity Framework Core, Clean Architecture, SOLID Principles, Middleware, Dependency Injection, Exception Handling.

Database: SQL Server, Query Optimization, Indexing, Stored Procedures, MongoDB (Basic), Database Design, Performance Optimization.

Testing & Debugging: Manual Testing, API Testing, Unit Testing, Debugging, Bug Fixing, Postman, Swagger/OpenAPI, Chrome DevTools

Cache & Infra: Redis (Caching, Session, Pub/Sub), Docker, Docker Compose, AWS S3, CI/CD, GitHub Actions.

AI & Tools: Retrieval-Augmented Generation (RAG), LLM Integration (API Key, Ollama), Vector Embeddings, SSE Streaming, Prompt Engineering, Postman, Swagger/OpenAPI, Git, Agile/Scrum.

PROFESSIONAL EXPERIENCE

Associate Software Engineer (Full Stack) - Nova Techset Pvt. Ltd., Chennai, India Jun 2022 - Present

Progression: Trainee → Junior → Associate Software Engineer

- Built and maintained 5+ enterprise web applications using Angular, ASP.NET Core, SQL Server, Docker, and Redis, supporting stable production deployments with 99.9% uptime.
- Improved backend performance by 15% by implementing Redis caching, optimizing SQL queries, adding database indexes, and making application-level improvements.
- Reduced Angular page load times by 20% using lazy loading, OnPush change detection, RxJS, and component-level performance analysis.
- Developed secure REST APIs using ASP.NET Core with JWT authentication and role-based access control (RBAC).
- Added real-time collaboration features using SignalR for live multi-user editing and synchronized workflows in publishing applications.
- Containerized applications using Docker and Docker Compose to keep development, testing, and production environments consistent.
- Integrated AWS S3 for secure document storage using presigned URLs with controlled access and URL expiration.
- Developed an AI-based chat assistant using Server-Sent Events (SSE), LLM API integration, and Ollama to provide real-time responses within enterprise applications.
- Created CI/CD pipelines using GitHub Actions to automate application builds, testing, and Docker image deployments.
- Participated in code reviews, developed reusable Angular components, and supported junior developers with implementation and troubleshooting.

KEY PROJECTS

1. Vendor Management System (VMS) - Publishing Workforce Platform

Stack: Angular 19, TypeScript, Tailwind CSS, .NET Core 8, C#, SQL Server, Redis, Docker, JWT, REST API, AI Assistant

- Built and maintained a full-cycle publishing workforce platform used by 300+ active freelancers, including copyeditors, typesetters, proofreaders, and DTP specialists, with onboarding and qualification workflows.
- Developed a competency assessment system for publishing-related skills with real-time scoring and automated result tracking, reducing vendor evaluation time from days to hours.

- Implemented role-based file access using JWT claims so that manuscript files and editorial assets were available only to qualified vendors after completing the required assessments.
- Developed invoice and billing workflows to calculate payments based on word counts, deliverable types, and agreed rates, reducing manual billing work.
- Added Redis caching for frequently used job listing and vendor lookup APIs and used Docker to containerize the application for consistent deployments.
- Integrated an AI-based chat assistant to provide contextual guidance to vendors during onboarding and reduce the need for manual support.

2. NEDIT - Enterprise Web-Based Content Editing & XML Proofing Platform

Stack: Angular 17, TypeScript, TinyMCE, ICE Plugin, SignalR, .NET Core, SQL Server, XML Processing, REST API

- Worked on an enterprise content editing and proofing platform for publishing workflows, covering content management, XML validation, and workflow automation.
- Developed rich-text editing features using TinyMCE, including text formatting, paragraph styles, footnotes, and dynamic image replacement for structured publishing workflows.
- Implemented change tracking using the ICE plugin to maintain revision history, compare content changes, and support proofing activities for distributed editorial teams.
- Implemented real-time multi-user collaboration using SignalR, allowing multiple users to edit content simultaneously and receive live updates to workflow changes.
- Developed online mathematical content editing with support for complex mathematical expressions, along with spell checking and XML validation to improve publishing accuracy.
- Maintained XML structure and formatting throughout the proofing workflow, from content ingestion through final publication, ensuring the output remained valid and consistent.

3. nPMS - Enterprise Project Management System

Stack: Angular 20, .NET Core 10, MongoDB, JWT, REST API, AI Integration

- Developed and maintained the frontend, backend, and AI integrations for nPMS, an enterprise project management platform with role-based login and access control for project teams.
- Built an automated issue-tracking and alert system for project managers handling 100+ issues per day. The system identifies at-risk or missed article issues and sends real-time notifications to help teams address them before delivery deadlines.
- Implemented role-based authentication and authorization (RBAC) to control access to project and issue-management modules based on user roles.

4. AI RAG Assistant - Retrieval-Augmented Generation Document Assistant

Stack: FastAPI, Python, Qdrant, Redis, Google Gemini API, Ollama, Docker, Render CI/CD

- Built a full-stack RAG (Retrieval-Augmented Generation) document assistant that supports PDF, DOCX, TXT, and CSV files and provides answers with source citations.
- Developed the document ingestion pipeline for text extraction, recursive chunking (1,000 tokens with 200-token overlap), and vector embedding generation. Stored embeddings in Qdrant and used cosine similarity search with top-k retrieval.
- Integrated Google Gemini API for LLM and embedding generation, using Gemini models for production and Ollama with bge-m3 and qwen2.5:7b for local development and testing.
- Used Redis for conversation history, document tracking, and response caching to reduce repeated LLM requests and improve response time.
- Containerized the application using Docker Compose and set up CI/CD deployment to Render.
- Developed the backend using FastAPI with dependency injection, making the chunking, embedding, and API layers easier to test using mocked external services.

Live: [AI-Rag-Agent](#)

EDUCATION

B.E. Computer Science - Alpha Engineering College, Anna University

Apr 2021

CGPA: 7.45 / 10

CERTIFICATIONS

- Skill Training Certificate - Web Developer, Government of Tamil Nadu, SCVT (Dec 2021)